Félix Gélinas

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Education

York University (Advisors: Nantel Bergeron and Vincent Pilaud) 2023 – expected August 2027
Doctor of Phylosophy in Mathematics (PhD) *Toronto, Ontario, Canada*

University of Toronto (Advisor: Swastik Kopparty) 2022 – 2023
Master of Science in Mathematics (MSc) *Toronto, Ontario, Canada*

Université du Québec à Montréal 2019 – 2022
Bachelor of Science in pure mathematics (BSc) *Montréal, Québec, Canada*

Research Experience

PhD project (Advisors: Nantel Bergeron and Vincent Pilaud) September, 2023 – Present
York University *Toronto, Ontario, Canada*

Master Thesis project (Advisor: Swastik Kopparty) May, 2023 – August, 2023
University of Toronto *Toronto, Ontario, Canada*

Summer internship in Research (Advisor: Hugh Thomas) May, 2022 – August, 2022
Université du Québec à Montréal *Montréal, Québec, Canada*

Summer internship in Research (Advisor: Hugh Thomas) May, 2021 – August, 2021
Université du Québec à Montréal *Montréal, Québec, Canada*

Publications

[1] Antoine Abram et al. "Ornamentation lattices and intreeval hypergraphic lattices". In: *Mathematische Zeitschrift* 314.1 (2026), p. 15. DOI: 10.1007/s00209-026-04080-x.

[2] Nicolas Andrews et al. "When are Hopf algebras determined by integer sequences?" In: *Journal of Pure and Applied Algebra* 230.9 (2026), p. 108326. ISSN: 0022-4049. DOI: 10.1016/j.jpaa.2026.108326.

[3] Gabriel Frieden, Félix Gélinas, and Étienne Soucy. "Cones and ping-pong in three dimensions". In: *Involve, a Journal of Mathematics* 17.1 (Mar. 2024), pp. 11–28. ISSN: 1944-4176. DOI: 10.2140/involve.2024.17.11.

Accepted for Publication

[1] Félix Gélinas, Adrien Segovia, and Hugh Thomas. *Proof of a conjecture of Matherne, Morales, and Selover on encodings of unit interval orders*. 2022. arXiv: 2212.12171 [math.CO]. URL: https://arxiv.org/abs/2212.12171.

Preprints

[1] Jose Bastidas et al. *Interval hypergraphic polytopes (or deformed associahedra), Tamari interval posets, and weeping willows*. 2026. arXiv: 2606.18376 [math.CO]. URL: https://arxiv.org/abs/2606.18376.

[2] Félix Gélinas and Yirong Yang. *Lattice characterization of cyclic interval hypergraphic posets*. 2026. arXiv: 2605.03913 [math.CO]. URL: https://arxiv.org/abs/2605.03913.

[3] Félix Gélinas. *Source characterization of the hypergraphic posets*. 2025. arXiv: 2508.16006 [math.CO]. URL: https://arxiv.org/abs/2508.16006.

Ongoing Projects

(1) Hypergraphic Polytopes
In collaboration with Jose Bastidas (LACIM), Vincent Pilaud (Universitat de Barcelona), Germain Poullot (Universität Osnabrück) and Eleni Tzanaki (University of Crete).

(2) Diameter of the Hypergraphic Polytopes
In collaboration with Germain Poullot (Universität Osnabrück).

(3) Hypergraphic canonical join complex
In collaboration with Louis Marin (Université Gustave Eiffel) and Yirong Yang (Georgia Institute of Technology).

Supervision of Summer Internship in Research

Catalan Hopf subalgebras of NCSym — May, 2024 – August, 2024
Supervising students Eric Schlums and Nicolas Andrews – York University — *Toronto, Ontario, Canada*

Using machine learning to find combiantorial bijections — May, 2024 – August, 2024
Supervising students Eniola Ogunjimi and Inderpreet Ganda – York University — *Toronto, Ontario, Canada*

Teaching Instructor (Teacher) at York University

Mathematics I for Biological and Health Sciences (MATH1506) — September 2026 – present

Teaching Assistant (TA) at York University

Grader

Mathematics I for Biological and Health Sciences (MATH1506) — *May 2026 – August 2026*
Introduction to Sets and Logic (MATH1190) — *January 2026 – April 2026*
Discrete Mathematics for Computer Science (MATH1019) — *January 2026 – April 2026*
Applied Calculus I (MATH1013) — *September 2025 – December 2025*
Discrete Mathematics for Computer Science (MATH1019) — *September 2025 – December 2025*
Introduction to Logic for Computer Science (MATH1090) — *May 2025 – August 2025*
Introduction to Logic for Computer Science (MATH1090) — *January 2025 – April 2025*
Differential Calculus with Applications (MATH1300) — *September 2024 – December 2024*
Introduction to Logic for Computer Science (MATH1090) — *September 2024 – December 2024*
Applied Multivariate and Vector Calculus (MATH2015) — *May 2024 – June 2024*
Integral Calculus with Applications (MATH1310) — *January 2024 – April 2024*
Differential Calculus with Applications (MATH1300) — *September 2023 – December 2023*

Tutorial Leader

Problems, Conjectures and Proofs (MATH1200) — *January 2025 – April 2025*
Problems, Conjectures and Proofs (MATH1200) — *September 2024 – December 2024*
Problems, Conjectures and Proofs (MATH1200) — *January 2024 – April 2024*
Problems, Conjectures and Proofs (MATH1200) — *September 2023 – December 2023*

Teaching Assistant (TA) at University of Toronto

Tutorial Leader and Grader

Calculus II (MAT136) — *January 2023 – April 2023*
Calculus I (MAT135) — *September 2022 – December 2022*

Teaching Assistant (TA) at Université du Québec à Montréal

Tutorial Leader

Calculus of Differential Forms (MAT2410) — *January 2022 – April 2022*

Organization

Co-organizer for Participants-Seminar of the CGGC October, 2025 – November 2025
Universitat de Barcelona (CRM) *Barcelona, Spain*

Other Experience

Referee for *Combinatorial Theory Journal* 2026
Moderator for Mathematical Summer in Paris July, 2021 – July 2021
Université Paris-Saclay *Online*

Presentations (Talks)

University of Washington Combinatorics Seminar April, 2026
University of Washington *Seattle, Washington, USA*
Séminaires de mathématiques discrètes December, 2025
Université de Liège *Liège, Liège, Belgique*
Combinatorial Geometries and Geometric Combinatorics November, 2025
Universitat Autònoma de Barcelona *Barcelona, Cataluña, Spain*
CGGC25 Participants Seminar October, 2025
Universitat Autònoma de Barcelona *Barcelona, Cataluña, Spain*
Canadian Discrete and Algorithmic Mathematics Conference (CanaDAM) May, 2025
University of Ottawa *Ottawa, Ontario, Canada*
Algebraic and enumerative combinatorics seminar May, 2025
University of Waterloo *Waterloo, Ontario, Canada*
Winter School and Workshop on Polytopes February, 2025
Universität Osnabrück *Osnabrück, Lower Saxony, Germany*
Séminaire du LACIM February, 2025
Université du Québec à Montréal (UQAM) *Montréal, Québec, Canada*
Combinatorial Algebra meets Algebraic Combinatorics (CAAC) January, 2025
The Fields Institute *Toronto, Ontario, Canada*
The Algebraic Combinatorics Workgroup October-November, 2024
The Fields Institute *Toronto, Ontario, Canada*
The Algebraic Combinatorics Workgroup May, 2024
The Fields Institute *Toronto, Ontario, Canada*
Graduate student Group May, 2024
The Fields Institute *Toronto, Ontario, Canada*
The Algebraic Combinatorics Workgroup September, 2023
The Fields Institute *Toronto, Ontario, Canada*
The Graduate Student Seminar UofT March, 2023
University of Toronto *Toronto, Ontario, Canada*
The Algebraic Combinatorics Workgroup March, 2023
The Fields Institute *Toronto, Ontario, Canada*
Applied Algebra Seminar March, 2023
York University *Toronto, Ontario, Canada*
The Graduate Students Seminar (UQAM) August, 2022
Université du Québec à Montréal (UQAM) *Montréal, Québec, Canada*
Pan-Québec colloquium for students of the ISM June, 2022
Université Laval *Québec, Québec, Canada*

Presentations (Poster)

Escuela Costarricense de Combinatoria	July 2026
Universidad de Costa Rica	*San José, San José, Costa Rica*
AlCoVe : an Algebraic Combinatorics Virtual Expedition	June, 2026
	Online Conference
ISM Discovery School: Combinatorics of Moduli of Curves	May, 2026
Université du Québec à Montréal (UQAM)	*Montréal, Québec, Canada*
SageMath Research Summer School (Sage Days 130.5)	May, 2026
Université du Québec à Montréal (UQAM)	*Montréal, Québec, Canada*
Combinatorial Algebra meets Algebraic Combinatorics (CAAC)	January, 2026
Dalhousie University	*Halifax, Nova Scotia, Canada*
Au-delà du Permutoèdre et de l'associaèdre	December, 2025
CIRM (Centre International De Rencontres Mathématiques)	*Marseille, Provence-Alpes-Côte d'Azur, France*
CGGC25 Closing Conference	November, 2025
Universitat Autònoma de Barcelona	*Barcelona, Cataluña, Spain*
AlCoVe : an Algebraic Combinatorics Virtual Expedition	May, 2025
	Online Conference
VIII Encuentro Colombiano de Combinatoria (ECCO)	June, 2024
Universidad del Cauca	*Popayàn, Cauca, Colombia*
Combinatorial Algebra meets Algebraic Combinatorics (CAAC)	January, 2024
Université du Québec à Montréal	*Montréal, Québec, Canada*
Combinatorial Algebra meets Algebraic Combinatorics (CAAC)	January, 2023
University of Waterloo	*Waterloo, Ontario, Canada*

Awards & Honors

Canada Graduate Research Scholarship – Doctoral program	2026-2029
NSERC– 120 000$	
Ontario Graduate Scholarship	2026 (Declined)
York University – 15 000$	
Academic Excellence Fund	2025
York University – 1 600$	
FGS Tuition Fee Bursary	2025
York University – 175$	
Kingsley Abu Imonikhe Graduate Award	2025
York University – 500$	
Academic Excellence Fund	2025
York University – 1 080$	
Vernon Oliver Stong Graduate Scholarships in Science	2024
York University – 10 000$	
Graduate Studies	2023-2027
York University – 112 964 $	
Graduate Studies	2023-2024
York University – 3 000$	
Graduate Studies	2022-2023
University of Toronto – 22 000$	

Undergraduate Student Research Awards 2022
NSERC – 6 000$

Advisor contribution NSERC 2022
Hugh Thomas – 1 500$

Supplements of the NSERC Undergraduate Student Research Awards 2022
FRQNT – 1 500 $

Undergraduate summer scholarship 2022 (Declined)
Institute of Mathematical Sciences (ISM) – 2 500 $

Undergraduate summer scholarship 2021
Institute of Mathematical Sciences (ISM) – 2 500 $

Advisor contribution ISM 2021
Hugh Thomas – 2 500$

Advisor contribution ISM 2021
Christophe Reutenauer – 2 500$

Summer Schools

Escuela Costarricense de Combinatoria (ECCO) 2026
CIMPA Research School, San José, San José, Costa Rica

ISM Discovery School: Combinatorics of Moduli of Curves 2026
S. Backman – M. Gillespie – J. Levinson – A. Morales, Montréal, Québec, Canada

SageMath Research Summer School (Sage Days 130.5) 2026
J. Bastidas – J.-P. Labbé – N. Lafrenière – A. Morales – X. Provençal – F. Saliola, Montréal, Canada

VIII Encuentro Colombiano de Combinatoria (ECCO) 2024
CIMPA Research School, Popayàn, Cauca, Colombia

ISM Discovery School: Advances in Geometric Combinatorics and Representation Theory 2024
R. González D'Léon – A. H. Morales – F. Saliola – H. Thomas, Montréal, Québec, Canada

Winter Schools

Winter School and Workshop on Polytopes 2025
M. Juhnke – M. Reitzner – C. Thäle , Osnabrück, Lower Saxony, Germany

Intensive Research Program / Workshop

Combinatorial Geometries and Geometric Combinatorics 2025
Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain

Lattice Theory Workshop 2025
The Banff International Research Station (BIRS), Banff, Alberta, Canada

Seminars & Conferences attended

University of Washington Combinatorics Seminar 2026
University of Washington, Seattle, Washington, USA

Algebraic and Enumerative Combinatorics Seminar of the University of Waterloo 2026
Karen Yeats, Waterloo, Ontario, Canada

LACIM seminar 2026
M. Garcia – S. Giraudo – C. Reutenauer – T. Yu, Montréal, Québec, Canada

Combinatorial Algebra meets Algebraic Combinatorics (CAAC) 2026
G. Denham – S. Faridi – S. Kara – H. Thomas – D. Veer, Halifax, Nova Scotia, Canada

Séminaires de mathématiques discrètes 2025
Université de Liège, Liège, Liège, Belgique

Nikolaus Conference 2025
RWTH Aachen University, Aachen, North Rhine-Westphalia, Germany

Au-delà du Permutoèdre et de l'associaèdre 2025
C. Ceballos – V. Pilaud – V. Pons – I. Rasskin, Marseille, Provence-Alpes-Côte d'Azur, France

Closing conference of CGGC 2025
Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain

CGGC 2025
K. Knauer – A. de Mier – A. Padrol – J. Pfeifle – V. Pilaud, Barcelona, Catalonia, Spain

CGGC25 Participants Seminar 2025
F. Gélinas – L. Jiang – L. Saud, Barcelona, Catalonia, Spain

Research school of CGGC - Matroid week 2025
Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain

Research school of CGGC - Polytope week 2025
Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain

Recollections on polyhedral geometry and (oriented) matroids 2025
Universitat de Barcelona (CRM), Barcelona, Catalonia, Spain

Seminar on Pure Mathematics 2025
The Hong Kong University of Science & Technology, New Territories, Sai Kung, Hong Kong

AlCoVe : an Algebraic Combinatorics Virtual Expedition 2025
L. Colmenarejo – M. Gillespie – O. Pechenik – L. Solus – F. Tom – L. Vecchi, Online Conference

Canadian Discrete and Algorithmic Mathematics Conference (CanaDAM) 2025
Ottawa, Ontario, Canada

Algebraic and Enumerative Combinatorics Seminar of the University of Waterloo 2025
Karen Yeats, Waterloo, Ontario, Canada

Combinatorics Seminar of UofT 2025
Vasu Tewari, Toronto, Ontario, Canada

Winter School and Workshop on Polytopes 2025
M. Juhnke – M. Reitzner Osnabrück – C. Thäle, Lower Saxony, Germany

LACIM seminar 2025
J. Bastidas – S. Giraudo – C. Reutenauer, Montréal, Québec, Canada

Combinatorial Algebra meets Algebraic Combinatorics (CAAC) 2025
S. Faridi – S. Kara – R. Orellana – G. Smith – V. Tewari – M. Zabrocki, Toronto, Ontario, Canada

The Algebraic Combinatorics Workgroup 2022-2026
Nantel Bergeron – Kelvin Chan – Vasu Tewari, The Fields Institute, Toronto, Ontario, Canada

Applied Algebra Seminar 2022-2025
Lucas Gagnon, York, Ontario, Canada

Workshop on Computational and Applied Enumerative Geometry 2024
T. Brysiewicz – L. Garcia Puente – N. Hein – J. Lopez Garcia – K. Purbhoo, Toronto, Ontario, Canada

Combinatorial Algebra meets Algebraic Combinatorics (CAAC) 2024
S. Brauner – J. Levinson – A. Morales – G. Park – F. Saliola – H. Thomas, Montréal, Québec, Canada

The Graduate Student Seminar UofT 2023
Graduate Student of UofT, Toronto, Ontario, Canada

Matroids - Combinatorics, Algebra and Geometry Seminar 2023
Ahmed Ashraf, The Fields Institute, Toronto, Ontario, Canada

Combinatorial Algebra meets Algebraic Combinatorics (CAAC) 2023
Sara Faridi–Selvi Kara –Oliver Pechenik –Matthew Satriano, Waterloo, Ontario, Canada

The Graduate Students Seminar (UQAM) 2022
Université du Québec à Montréal (UQAM), Montréal, Québec, Canada

Non-Commutative Algebra, Representation Theory, Special Functions (Conference) 2022
CRM, Montréal, Québec, Canada

Pan-Quebec colloquium for students of the ISM 2022
Institute of Mathematical Sciences (ISM), Université Laval, Québec, Québec, Canada

Open Problems in Algebraic Combinatorics (OPAC) 2022
University of Minessota, Minneapolis, Minessota, USA

Graduate Student Conference on Combinatorics, Commutative Algebra and Geometry 2022
University of Minessota, Minneapolis, Minessota, USA

LACIM seminar 2021-2022
LACIM, Montréal, Québec, Canada